

Part G: Construction of a Testable Prototype

Once we had solidified the design requirements for our 3D-printed tree stand, we moved forward with the development of a functional prototype that would allow for thorough testing and evaluation. The purpose of this prototype was to assess both the structural integrity and practical usability of the tree stand in real-world scenarios, as well as to identify any necessary improvements before final production.

Final Design Sketches

We began by refining our design sketches, ensuring that each component of the tree stand was optimized for weight reduction, strength, and easy assembly. Our primary focus during this stage was to minimize material usage without sacrificing safety and load capacity. This led us to experiment with lattice structures and strategic reinforcement points, inspired by principles of structural engineering to distribute weight effectively.

The design also considered the slight flex inherent in the 3D-printed material (PLA with carbon fiber reinforcement), which, although initially seen as a limitation, was turned into an advantage for shock absorption and enhanced grip when mounted on tree bark. This decision was made after evaluating other designs and materials, which either lacked flexibility or were too heavy for practical use.

3D Modeling and Printing

Using advanced CAD software, we constructed a detailed 3D model of the tree stand, complete with modular segments for easy transport and assembly in the field. We incorporated mounting points, safety latches, and adjustable straps directly into the model to ensure a secure fit during use.

The prototype was printed using a carbon-fiber-reinforced PLA filament, chosen for its excellent strength-to-weight ratio and its durability under load. We also opted for a 50% infill density with a triangular lattice structure to enhance structural integrity while minimizing weight. The final prototype weighed approximately 35% less than traditional metal-based stands.

We performed preliminary assembly tests to confirm that all components fit as intended and to identify any points of failure before field testing began. The slight flex observed during assembly was within acceptable ranges, contributing to better grip and minor shock absorption.